

Messier catalog

A catalog of nebulae (including some objects now known to be galaxies) compiled, and published in 1781, by French astronomer Charles Messier and his assistant Pierre Méchain. Objects in this catalog are assigned an individual number, preceded by "M". For example, the Andromeda galaxy is M31. See also *New General Catalog*

Meteor

The short-lived streak of light or "shooting star" seen when a small Solar System body burns up on entering Earth's atmosphere. If the body survives to reach the ground it is termed a meteorite.

Meteorite

A small solid Solar System body that has survived passing through the atmosphere of Earth or another planet and has reached the ground.

Microwave radiation

Electromagnetic radiation with wavelengths shorter than radio waves but longer than infrared and visible light. See also *electromagnetic radiation*, *electromagnetic spectrum*

Milky Way

(1) Originally, the luminous band across the night sky that represents the combined light of vast numbers of stars and nebulae in the disk of our home galaxy. (2) Now used as a name for the galaxy itself.

Mira variable

A class of giant variable stars whose brightness varies over a period of around 100 to 500 days. See also *variable star*

Moon

A natural satellite orbiting a planet. Spelled with a capital, "the Moon" refers to the Earth's moon.

Multiple star

A system consisting of three or more stars bound together by gravity and orbiting around one another. See also *binary star*

N**NASA**

Short for National Aeronautics and Space Administration, the main space agency of the United States.

Nebula

A cloud of gas and dust in interstellar space. Some nebulae are sites of star formation, while others are produced at the end of a star's life. See also *planetary nebula*

Neutrino

A particle of exceedingly low mass and zero electrical charge that travels close to the speed of light and rarely interacts with other matter.

Neutron star

An extremely dense compact star made of tightly packed neutrons (neutral subatomic particles). Neutron stars are formed by supernova explosions not massive enough to create a black hole. See also *pulsar*

New General Catalog (NGC)

A catalog of star clusters and nebulae (including objects now known to be galaxies) compiled by J.L.E. Dreyer in 1888. Objects are assigned an individual number, preceded by "NGC". With amendments, this system is still in use today. See also *Messier catalog*

Nova

A star that suddenly brightens, then fades back to its original brightness over a period of weeks or months. The brightening happens when a fusion reaction is triggered on the surface of a white dwarf star by gas flowing from another star. See also *fusion*, *supernova*

Nuclear fusion

See *fusion*

O**Observable Universe**

That part of the Universe from which light has had time, since the Big Bang, to reach Earth.

Oort cloud

A spherically distributed collection of trillions of icy bodies such as cometary nuclei that is believed to surround the Solar System and extend more than 1 light-year distant from the Sun.

Open cluster

A relatively spread-out cluster of stars that all formed at the same time. See also *globular cluster*

Optical double

See *double star*

Orbit

The path of an astronomical body when it is revolving around another under the influence of gravity.

P**Particle**

In astronomical contexts this usually means a subatomic particle, such as a proton or neutron, or more exotic particles of similarly tiny size.

Photosphere

The layer of the Sun or other star from which most of the light is emitted and that forms its visible surface. See also *chromosphere*, *corona*

Planet

A large body orbiting a star. A planet is sufficiently massive for its gravity to have formed it into a round shape and also for it to have cleared its orbital path of other objects. See also *dwarf planet*

Planetary nebula

A glowing shell of gas ejected by a star of similar mass to the Sun when coming to the end of its life. The term was first used by William Herschel for circular nebulae that looked similar to a planet.

Plasma

A mixture of electrons and positive ions that behaves like a gas, but conducts electricity and is affected by magnetic fields. The Sun and other stars are made of hot plasma. See also *ion*

Precession

A slow cyclic change in the direction of Earth's axis (i.e. the direction in which the north and south poles "point"), which takes 25,800 years to complete. A similar kind of movement is seen in a spinning top. The term is also applied to other astronomical cycles, such as the slow change in position of the farthest point of a planet's orbit.

Prominence

A huge eruption of glowing plasma into the Sun's corona, often in a looping shape. See also *corona*, *plasma*

Protoplanet

A precursor of a planet, which develops through the gradual aggregation of smaller bodies in the protoplanetary disk that forms around many new stars. Planets are thought to form by collision of protoplanets.

Protostar

A star in the early stages of formation, before hydrogen fusion has begun.

Pulsar

A rapidly rotating neutron star that is sending out powerful jets of radiation from its magnetic poles. Pulses are detected if the jets happen to sweep by in Earth's directions as the neutron star spins. See also *neutron star*.

Pulsating variable

See *variable star*.

Q**Quasar**

A compact but extremely powerful source of radiation, now believed to be a type of highly luminous active galaxy. Most quasars are at extreme distances from our own galaxy and we are observing them as they were early in the history of the Universe. See also *active galaxy*

R**Radio telescope**

An instrument that is designed to detect radio waves from astronomical sources. The most familiar type is a concave dish that collects radio waves and focuses them onto a detector.

Red dwarf star

A cool, red, low-luminosity star. Red dwarfs are common in the Universe and are very long-lived.

Red giant star

A greatly expanded reddish star with a low surface temperature that forms at the end of the life of Sun-like stars. It is "giant" in its size and luminosity, rather than in mass. See also *super giant*

Redshift

The shifting to a lower frequency of electromagnetic radiation when it comes from an object moving away from an observer. It can be compared to the siren on an emergency vehicle that sounds a lower note when speeding away.

Reflector

A telescope in which the light is collected and focused by a curved mirror. See also *refractor*.

Refractor

A telescope in which the light is collected and focused by a lens. See also *reflector*.

Relativity

Two theories developed in the early 20th century by Albert Einstein. The special theory of relativity describes how the relative motion of observers affects their measurements of mass, length, and time. One consequence is that mass and energy are equivalent. The general theory of relativity treats gravity as a distortion of spacetime. See also *spacetime*

Right ascension

The equivalent on the celestial sphere of longitude on Earth. The right ascension of a star is its angular distance east of a point in the sky called the first point of Aries. It is expressed in hours, minutes, and seconds, 1 hour being the equivalent of 15 degrees. See also *declination*